LÊ THẾ VINH

AUTOMATION ARCHITECT

05/07/2006

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56, TX21, Thạnh Xuân Ward, District 12

0966013783

Skills

English Proficiency

Professional Traits

Interpersonal Skills

Analytical Thinking

Technical Foundation

Collaboration

Certificates

IELTS | **2022**

IELTS 6.0

COMPUTATIONAL THINKING FOR PROBLEM SOLVING OF PENNSYLVANIA UNIVERSITY | **2024**

Prizes

THIRD PRIZE IN MATHEMATICS, GROUP V OF THE OLYMPIC 30/4 COMPETITION.

2022

Introduction

As a dedicated Automation Architect with extensive experience in designing and delivering enterprise-scale automation solutions, I am driven by the goal of transforming complex business processes into efficient, intelligent, and scalable systems. My expertise lies in building robust automation frameworks, aligning technology strategies with business objectives, and ensuring high standards of reliability, security, and performance. I am passionate about leveraging modern technologies such as cloud platforms, AI-powered automation, and DevOps practices to create sustainable digital ecosystems. With a proactive mindset and strong leadership capabilities, I continuously strive to innovate, optimize, and guide teams in solving complex technical challenges while delivering measurable business value.

Experience

PROGRAMMING TEACHER

ALGORITHMICS | **2024 - current**

- Assess students' capabilities through an entrance test and analyze their strengths/weaknesses to develop a personalized learning roadmap.
- Prepare detailed lesson plans, combining fundamental and advanced exercises, and self-compiled practice tests for each topic.
- Deliver clear instruction, illustrate concepts with real-world examples, and guide students on quick and effective problem-solving methods.
- Support students with regular practice, thoroughly correct mistakes, and provide encouraging feedback.
- Monitor progress through periodic reports: focusing on accuracy, solution speed, and test results; adjusting teaching methods promptly.
- Organize mock exams, analyze the test matrix, and provide guidance on time management strategies during the examination.
- Maintain close communication with parents, report on progress, and propose at-home support options.
- Update traditional teaching methods, incorporating digital tools when necessary, but prioritizing effective 1:1 interaction.

ACHIEVED 9TH PLACE (TEAM CATEGORY) AT THE WECODE CHALLENGE WITHIN THE UIT BOOTCAMP.

2025

ACHIEVED 6TH PLACE AT THE IT ARENA COMPETITION.

2025

ACHIEVED 5TH PLACE AT THE CURSOR HACKATHON HCMC.

2025

Education

INFORMATION SYSTEM

VNUHCM - UNIVERSITY OF INFORMATION TECHNOLOGY | **2024 - current**

Current GPA: 3.1

UIT ALGO SUMMER BOOTCAMP 2025

Activities

UIT ALGO AUTUMN BOOTCAMP 2025

Collaborator | **9/2025 - 10/2025**

Projects

DEVELOPED A VEHICLE RE-IDENTIFICATION (RE-ID) MODEL USING THE YOLOV8 FRAMEWORK.

04/2025 - 05/2025

- **Duration:** 04/2025 – 05/2025

Description:
A computer vision system designed to automatically detect lane markings and traffic signal states in real-time to support intelligent traffic monitoring and automation.

Role:
Computer Vision Engineer

Responsibilities:

- Developed and trained deep learning models for lane line segmentation and traffic light detection.
- Built automated data preprocessing and annotation workflows to improve dataset quality and training efficiency.
- Integrated detection modules into a real-time processing pipeline for video stream analysis.
- Optimized inference performance to ensure stable real-time deployment.

Technologies:
Python, YOLOv8, OpenCV, PyTorch/TensorFlow, NumPy

TRAFFIC LANE LINE AND SIGNAL LIGHT DETECTION

| 05/2025 - 07/20-25

- **Duration:** 05/2025 – Present
- **Description:** An optimized and lightweight detection model engineered to identify the location of lane lines and the state (Green/Red/Yellow) of traffic signals in real-time.
- **Role:** Computer Vision Technician
- **Responsibilities:**
  Employed the MobileNetV3 architecture to ensure high inference speed, making it suitable for edge devices.
  Conducted model training for both Object Detection and Semantic Segmentation (for lane lines).
  Optimized the model via Quantization to maintain an accuracy of over 90% while achieving fast processing speeds.
- **Technologies:** Python, MobileNetV3, TensorFlow Lite/OpenVINO (Deployment Optimization), Model Quantization Techniques.

TRAFFIC VIOLATION DATASET

| 06/2025 - 8/2025

- **Duration:** 06/2025 – Present
- **Description:** A pioneering project to build a comprehensive traffic violation dataset, coupled with the development of a real-time AI system for infraction detection.
- **Role:** Co-Author/Data Curator & Developer
- **Responsibilities:**
  Managed the process of data collection, quality control, and detailed annotation of complex traffic violation scenarios.
  Developed Object Detection and Tracking models, integrating them with business logic to accurately determine and classify violation behaviors.
  Goal: Build a high-accuracy system capable of providing timely alerts for violations.
- **Technologies:** Python, Annotation Tools (CVAT/LabelImg), YOLO/Faster R-CNN, Action Recognition Techniques.

SAFE JOURNEY | **11/2025 - current**

Description:
An intelligent traffic monitoring system designed to automatically detect vehicles and identify traffic violations in real-time to enhance road safety management.
Role:
AI Integration Developer
Responsibilities:

- Integrated object detection and tracking models into a real-time video processing pipeline.
- Designed automated workflows for violation detection and alert generation.
- Optimized model inference to ensure stable performance in real-world environments.
- Collaborated with the backend team to deploy and maintain system services.

Technologies:
Python, YOLOv8, DeepSORT, OpenCV, Flask/FastAPI, REST APIs

UIT ASSISTANT EXAM APP | 1/2025 - Present

Description:
An end-to-end MVP platform developed within 3 weeks to support UIT students with academic services and an AI-powered exam practice engine, featuring a RAG-based chatbot, structured exam bank, smart reminders, and automated data pipelines.
Role:
Technical Lead (Automation & AI Systems)
Responsibilities:

- Architected the overall system design, defining module boundaries (Academic Services, RAG Assistant, Practice Engine) and establishing API contracts between frontend, backend, and data pipelines.
- Led a 9-member cross-functional team (Data & Tech), assigning roles, defining deliverables, and ensuring milestone completion within a 3-week MVP timeline.
- Designed and governed the standardized Question Bank schema (JSON → PostgreSQL), including validation rules, QC sampling strategy, and dataset versioning for maintainability.
- Built and supervised the automation pipeline for exam ingestion (PDF/Image/Word → OCR/Parsing → Normalization → QC → Database import).
- Designed and implemented the RAG architecture (document ingestion, chunking, embedding, vector retrieval, citation-based responses, and guardrails to prevent hallucination).
- Oversaw backend development using FastAPI, including endpoints for practice generation, attempt lifecycle (start/submit/score/history), search agent, and rule-based smart nudges triggered by exam schedules.
- Established CI/CD workflow, staging environment, logging strategy, and demo-readiness checklist to ensure system stability and scalable deployment.
- Coordinated cross-team integration, conducted technical reviews, and resolved architectural blockers to maintain delivery velocity.

Technologies:
Python, FastAPI, PostgreSQL, Qdrant, LangChain, React (Vite), Docker, Git, REST APIs